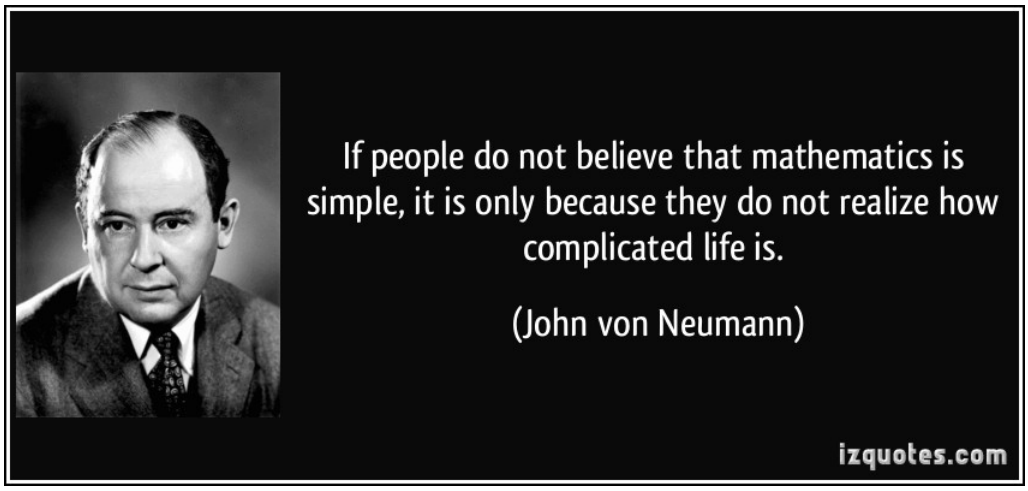
Chapter 3
Sequences of Real Numbers

Note 1

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INTRODUCTION

Sequences are a fundamental concept of utmost importance in analysis. Convergent sequences give us a systematic way of talking about particular real numbers which otherwise might be hard to describe. In this chapter, we will be concerned with sequences in $\mathbb{R}$ and will discuss what we mean by the convergence of these sequences.

3.1 : LIMIT OF SEQUENCES

- A **sequence of real numbers** (or a **sequence in** $\mathbb{R}$) is a function whose domain is $\mathbb{Z}^+$.
- A sequence f whose *terms* are $x_n := f(n)$ will be denoted by $x_1, x_2, \ldots$ or $\{x_n\}_{n\in\mathbb{Z}^+}$, or $\{x_n\}_{n=1}^{\infty}$, or $\{x_n\}$.
- Thus,

  (i). $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots$ represents the sequence $\left\{\frac{1}{2^{n-1}}\right\}_{n\in\mathbb{Z}^+}$

  (ii). $-1, 1, -1, 1, \ldots$ represents the sequence $\{(-1)^n\}_{n\in\mathbb{Z}^+}$

- It is more important not to confuse a sequence $\{x_n\}_{n\in\mathbb{Z}^+}$ with the set $\{x_n : n \in \mathbb{Z}^+\}$; these are two entirely different concepts.

  (i). The sequences $1, 2, 3, 4, \ldots$ is different from $2, 1, 3, 4 \ldots$, but as sets $\{1, 2, 3, 4, \ldots\}$ is identical with $\{2, 1, 3, 4, \ldots\}$.

  (ii). The sequence $1, -1, 1, -1, \ldots$ is infinite, but the set $\{(-1)^n : n \in \mathbb{Z}^+\}$ has only two points.

The limit concept is one of the fundamental building blocks of analysis. Note that a sequence of real numbers $\{x_n\}$ converges to a number a if x_n gets near a (i.e., the distance between a and x_n gets small) as n gets large. Thus, given $\varepsilon > 0$ (**no matter how small**), if n is large enough, $|x_n - a|$ is smaller than ϵ. This leads us to a formal definition of the limit of a sequence.

Definition 3.1.1

A sequence of real numbers $\{x_n\}$ is said to **converge** to a real number $a \in \mathbb{R}$ if and only if for every $\varepsilon > 0$ there is an $N \in \mathbb{Z}^+$ (which is in general depends on ε) such that

$$n \geq N \quad \text{implies} \quad |x_n - a| < \varepsilon.$$

or equivalently

$$|x_n - a| < \varepsilon \quad \text{whenever} \quad n \geq N \ .$$

Here a is called the **limit** of the sequence $\{x_n\}$, and we write $\lim\limits_{n\to\infty} x_n = a$. If a sequence does not converge to a real number, it is said to **diverge**.

- We shall use the following phrases and notations interchangeably for $a = \lim\limits_{n\to\infty} x_n$:

  (i). $\{x_n\}$ converges to a;

  (ii). $x_n \to a$ as $n \to \infty$;

  (iii). the *limit* of $\{x_n\}$ exists and equals a.

- When $x_n \to a$ as $n \to \infty$, you can think of $\{x_n\}$ as a sequence of approximations to a, and ε as an upper bound for the error of these approximations.

- The number N in **Definition 3.1.1** is chosen so that the error is less than ε when $n \geq N$.

  In general, the smaller ε gets, the larger N must be.



Figure 3.1.1

- Notice by definition that $\{x_n\}$ converges to a **if and only if** $|x_n - a| \to 0$ as $n \to \infty$.

  In particular, $x_n \to 0$ **if and only if** $|x_n| \to 0$ as $n \to \infty$.

- According to **Definition 3.1.1**, to prove that a particular limit exists, given an arbitrary $\varepsilon > 0$, no matter how small, we must describe how to choose an N such that $n \geq N$, implies $|x_n - a| < \varepsilon$.

  In particular, ε is usually introduced BEFORE N is specified, and N often is defined to depend on ε. Since $|x_n - a| < \varepsilon$ for all $n \geq N$, N CANNOT depend on n.

Example 1

(i). Prove that $\dfrac{1}{n} \to 0$ as $n \to \infty$.

(ii). If $x_n \to 2$, prove that $\dfrac{(2x_n + 1)}{x_n} \to \dfrac{5}{2}$ as $n \to \infty$.

Solution :

(i).

- Let $\varepsilon > 0$. Use the **Archimedean Principle** to choose $N \in \mathbb{Z}^+$ such that $N > \dfrac{1}{\varepsilon}$.
- By taking the reciprocal of this inequality, we see that $n \geq N$ implies $\dfrac{1}{n} \leq \dfrac{1}{N} < \varepsilon$.
- Since $\dfrac{1}{n}$ are all positive, it follows that $\left|\dfrac{1}{n}\right| < \varepsilon$ for all $n \geq N$.

(ii).

- Let $\varepsilon > 0$. Since $x_n \to 2$, apply **Definition 3.1.1** to this $\varepsilon > 0$ to choose $N_1 \in \mathbb{Z}^+$ such that $n \geq N_1$ implies $|x_n - 2| < \varepsilon$.
- Next, apply **Definition 3.1.1** with $\varepsilon = 1$ to choose N_2 such that $n \geq N_2$ implies $|x_n - 2| < 1$.

- By the **Fundamental Theorem of Absolute Values**, we have $n \geq N_2$ implies $x_n > 1$ (i.e., $2x_n > 2$).
- Set $N = \max\{N_1, N_2\}$ and suppose that $n \geq N$.
- Since $n \geq N_1$, we have $|2 - x_n| = |x_n - 2| < \varepsilon$.
- Since $n \geq N_2$, we have $0 < \dfrac{1}{2x_n} < \dfrac{1}{2} < 1$.
- It follows that
$$\left|\frac{2x_n + 1}{x_n} - \frac{5}{2}\right| = \frac{|2 - x_n|}{2x_n} < \frac{\varepsilon}{2x_n} < \varepsilon$$
for all $n \geq N$.

The following two results show that a given sequence can have no limits or one limit, but no more.

Example 2 The sequence $\{(-1)^n\}_{n\in\mathbb{Z}^+}$ has no limit.

Solution :

- Suppose that $(-1)^n \to a$ as $n \to \infty$ for some $a \in \mathbb{R}$.
- Given $\varepsilon = 1$, $\exists\ N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|(-1)^n - a| < \varepsilon$.
- For n odd this implies $|1 + a| = |-1 - a| <$, and for n even this implies $|1 - a| < 1$.
- Hence, we have
$$\begin{aligned} 2 &= |1 - a + a + 1| \\ &\leq |1 - a| + |1 + a| < 1 + 1 = 2; \end{aligned}$$
that is, $2 < 2$, which is a contradiction.

Remark 1 : When exists, limit of a sequence is unique.

Proof of Remark 1.

- Suppose that $\{x_n\}$ converges to two limits, a and b.
- By definition, given $\varepsilon > 0$, $\exists\ N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|x_n - a| < \dfrac{\varepsilon}{2}$ and $|x_n - b| < \dfrac{\varepsilon}{2}$.

- Thus it follows from the triangle inequality that

$$|a-b| \leq |a-x_n| + |x_n - b| < \varepsilon;$$

that is, $|a-b| < \varepsilon$ for all $\varepsilon > 0$.

- We conclude, by **Theorem 2.3** in **Section 2.2**, that $a = b$.

□

Definition 3.1.2

By a **subsequence** of a sequence $\{x_n\}_{n\in\mathbb{Z}^+}$, we shall mean a sequence of the form $\{x_{n_k}\}_{k\in\mathbb{N}}$, where each $n_k \in \mathbb{Z}^+$ and $n_1 < n_2 < \ldots$

- Thus a subsequence $x_{n_1}, x_{n_2}, \ldots$ of $x_1, x_2, \ldots$ is obtained by "deleting" from $x_1, x_2, \ldots$ all x_n's except those such that $n = n_k$ for some k. For example,

  (i). $1, 1, \ldots$ is a subsequence of $(-1)^n$ obtained by deleting every other term (set $n_k = 2k$).

  (ii). $\frac{1}{2}, \frac{1}{4}, \ldots$ is a subsequence of $\frac{1}{n}$ obtained by deleting all nondyadic fractions; that is, deleting $\frac{1}{3}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \ldots$ (set $n_k = 2^k$).

- Subsequences are sometimes used to correct a sequence that behaves badly or to speed up convergence of another that converges slowly. For example,

  (i). $\left\{\frac{1}{n}\right\}$ converges much more slowly to zero than its subsequence $\left\{\frac{1}{2^n}\right\}$.

  (ii). $\{(-1)^n\}$ does not converge at all (See **Example 2** above), but its subsequence $1, 1, \ldots$ converges to 1 immediately.

- If $x_n \to a$ as $n \to \infty$, then the x_n's get near a as n gets large. Since n_k gets large as k does, it comes as no surprise that any subsequence of a convergent sequence also converges.

Remark 2 : If $\{x_n\}_{n\in\mathbb{Z}^+}$ converges to a and $\{x_{n_k}\}_{k\in\mathbb{Z}^+}$ is any subsequence of $\{x_n\}_{n\in\mathbb{Z}^+}$, then x_{n_k} converges to a as $k \to \infty$.

Proof of Remark 2.

- Let $\varepsilon > 0$ and choose $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|x_n - a| < \varepsilon$.
- Since $n_k \in \mathbb{Z}^+$ and $n_1 < n_2 < \ldots$, by Mathematical Induction on $k \in \mathbb{Z}^+$ it follows that $n_k \geq k$.
- Hence, $k \geq N$ implies $|x_{n_k} - a| < \varepsilon$; that is, $x_{n_k} \to a$ as $k \to \infty$.

□

The following concepts also play an important role in the theory of sequences.

Definition 3.1.3

Let $\{x_n\}$ be a sequence of real numbers.

(i). The sequence $\{x_n\}$ is said to be *bounded above* if and only if the set $\{x_n : n \in \mathbb{Z}^+\}$ is bounded above.

(ii). The sequence $\{x_n\}$ is said to be *bounded below* if and only if the set $\{x_n : n \in \mathbb{Z}^+\}$ is bounded below.

(iii). $\{x_n\}$ is said to be *bounded* if and only if it is bounded both above and below.

Note :

- Combining **Definition 3.1.3** and **Definition 2** in Section 2.3, we see that $\{x_n\}$ is bounded above (respectively, below) if and only if there is an $M \in \mathbb{R}$ such that $x_n \leq M$ for all $n \in \mathbb{Z}^+$ (respectively, if and only if there is an $m \in \mathbb{R}$ such that $x_n \geq m$ for all $n \in \mathbb{Z}^+$).
- It is easy to check (see **Exercise 5**) that $\{x_n\}$ is bounded if and only if there is a $C > 0$ such that $|x_n| \leq C$ for all $n \in \mathbb{Z}^+$. In this case we shall say that $\{x_n\}$ is *bounded,* or *dominated*, by C.

Theorem 3.1 : Every Convergent sequence is bounded.

Proof of Theorem 3.1.

- Given $\varepsilon = 1$, there is an $N \in \mathbb{Z}^+$ such that $n \geq N$ implies $|x_n - a| < 1$.
- Hence by the triangle inequality, $|x_n| < 1 + |a|$ for all $n \geq \mathbb{Z}^+$.
- On the other hand, if $1 \leq n \leq N$, then
$$|x_n| \leq M := \max\{|x_1|, |x_2|, \ldots, |x_N|\}.$$
- Therefore, $\{x_n\}$ is dominated by $\max\{M, 1 + |a|\}$.

□

Note : By Example 2, the converse of Theorem 3.1 is false.

EXERCISES

(1). Decide which of the following statements are true and which are false. Prove the true ones and provide a counterexample for the false ones.

(a). If $\{x_n\}$ converges, then $\left\{\frac{x_n}{n}\right\}$ also converges.

(b). If $\{x_n\}$ does not converge, then $\left\{\frac{x_n}{n}\right\}$ does not converge.

(c). If $\{x_n\}$ converges and $\{y_n\}$ is bounded, then $\{x_n y_n\}$ converges.

(d). If $\{x_n\}$ converges to zero and $y_n > 0$ for all $n \in \mathbb{Z}^+$,, then $\{x_n y_n\}$ converges.

(2). Using the method of **Example 1 (i)**, prove that the following limits exist.

(a). $2 - \frac{1}{n} \to 2$ as $n \to \infty$

(b). $1 + \frac{\pi}{\sqrt{n}} \to 1$ as $n \to \infty$

(c). $3\left(1 + \frac{1}{n}\right) \to 3$ as $n \to \infty$

(d). $\frac{2n^2+1}{3n^2} \to \frac{2}{3}$ as $n \to \infty$

(3). Suppose that $\{x_n\}$ is a sequence of real numbers that converges to 1 as $n \to \infty$. Using **Definition 3.1.1**, prove that each of the following limits exists.

(a). $1 + 2x_n \to 3$ as $n \to \infty$

(b). $\frac{(\pi x_n - 2)}{x_n} \to \pi - 2$ as $n \to \infty$

(c). $\frac{(x_n^2 - e)}{x_n} \to 1 - e$ as $n \to \infty$

(4). For each of the following sequences, find two convergent sub-sequences that have different limits.

(a). $3 - (-1)^n$

(b). $(-1)^{3n} + 2$

(c). $\frac{(n - (-1)^n n - 1)}{n}$

(5). **(a)**. Prove that $\{x_n\}$ is bounded if and only if there is a $C > 0$ such that $|x_n| \leq C$ for all $n \in \mathbb{Z}^+$.

(b). Suppose that $\{x_n\}$ is bounded. Prove that $\frac{x_n}{n^k} \to 0$, as $n \to \infty$, for all $k \in \mathbb{Z}^+$.